Concrete Antifer Cubes

Executive Summary

Antifer cubes are massive grooved unreinforced concrete armour units used in rubble-mound breakwaters and related maritime protection works. They were developed for the Port of Antifer, Normandy, in the early 1970s, as a robust improvement over the plain concrete cube. The geometry remains compact and structurally conservative, but adds four lateral grooves and controlled tapering to improve hydraulic behaviour, roughness, interlock and resistance to face-to-face rearrangement.

This repository contains a Python production tool that generates a closed vectorial Antifer armour-unit solid from an input unit weight `W` and concrete unit weight `Wc`. The tool exports the same validated geometry to IFC, STL, OBJ and DXF. It is a geometric CAD/BIM production and documentation tool; hydraulic stability design remains a separate project-specific task.

Two dimensional conventions are implemented.

Convention	Purpose	Volume relation	Reference-height meaning
Pita (1986)	historical proportional Antifer reference	V/H3 = 1.024700000000	drawing height used in the historical coefficient set
Carvalho (2026)	volume-normalised CAD/BIM reference	V/H3 = 1.000000000000	H = Dn = V1/3

The same vectorial topology is used in both conventions: a tapered body, no top bevel, four lateral plan chamfers and four single circular-arc side grooves. The Carvalho convention is obtained by horizontally normalising the Pita plan and groove coefficients by a common factor so that the final non-dimensional volume coefficient becomes unity.

Repository Scope

This repository is concerned with the reproducible geometric reconstruction of Antifer cubes. It does not replace hydraulic model testing, breakwater stability design, overtopping analysis, toe design, crest design or construction-method verification.

The main outputs are:

Output	Description
IFC	IFC4 `IfcFacetedBrep` model with property and quantity data
STL	closed triangular surface mesh in metres
OBJ	indexed triangular surface mesh in metres
DXF	AutoCAD 2018 / AC1032 file with an ACIS-backed `3DSOLID`
TXT	`output.txt` calculation and validation report

Historical and Engineering Context

The Antifer cube is an improved massive concrete cube. It should not be treated as a slender high-interlock unit. Its engineering value lies in combining hydraulic improvement with structural robustness.

The unit became especially relevant after the structural-integrity problems observed in some slender concrete armour units during the late 1970s and early 1980s. Antifer cubes retain large resistant concrete sections and avoid slender arms, reducing the risk of bending-sensitive prototype breakage. This robustness comes at the cost of relatively high concrete consumption compared with modern monolayer armour units.

Representative contexts in which Antifer units have been used or studied include:

Case	Main relevance
Port of Antifer	origin and prototype context
Sines West Breakwater	repair and rehabilitation after Dolos armour-layer failure
Zeebrugge	prototype overtopping and run-up measurements
Douro Bar / Cabedelo do Douro	model testing, high-density units and monitoring
Leixoes Outer Breakwater	contemporary Portuguese port-extension use
Lajes das Flores	Atlantic island-port emergency protection and reconstruction
Mohammedia	severe-exposure international breakwater context
Brunei	documented manufacture, curing and placement practice

These examples are engineering context only. They do not define a universal stability coefficient.

Geometry and Dimensional Convention

Adopted Production Geometry

Item	Adopted convention
Vertical reference	`H` is the total unit height
Bottom section	`A` is the bottom bounding width at $z = 0$
Top section	`B` is the top bounding width at $z = H$
Top face	single horizontal plane; no top bevel
Chamfers	`D` is a lateral plan-chamfer offset
Grooves	four identical side grooves
Groove type	one continuous circular arc per side
Groove radius	`R`, with $r = R/H$
Groove centre offset	`E`, with $e = E/H$, outside the side face
Generated groove depth	Carc = R - E; carc = r - e
Groove half-opening	Q = sqrt(R2 - E2); q = Q/H = sqrt(r2 - e2)

Item	Adopted convention
Geometry source	pure vectorial B-rep generation; no imported template geometry

Master Coefficient Table

All Pita and Carvalho coefficients used by the production script are consolidated here. Later sections refer back to this table instead of repeating the same coefficient set.

Quantity	Meaning	Pita (1986)	Carvalho (2026)
KV	final volume coefficient, V/H3	1.024700000000	1.000000000000
f	horizontal factor applied to Pita coefficients	1.000000000000	0.987874174182
a = A/H	bottom width coefficient	1.086000000000	1.072831353161
b = B/H	top width coefficient	1.005000000000	0.992813545052
d = D/H	lateral chamfer coefficient	0.024000000000	0.023708980180
r = R/H	groove-radius coefficient	0.121500000000	0.120026712163
e = E/H	outside groove-centre offset	0.025906459610	0.025592322393
carc = r - e	generated circular-arc groove-depth coefficient	0.095593540390	0.094434389770
q = sqrt(r2 - e2)	groove half-opening coefficient	0.118705961731	0.117266553915
2q	full groove-opening coefficient	0.237411923462	0.234533107831

The historical Pita notation often reports C/H = 0.095600. The production geometry uses the volume-consistent circular-arc value carc = 0.095593540390, which is the generated depth obtained from Carc = R - E.

Algebraic Volume Definition

The generated unit volume is computed from a simple geometric balance.

```
\frac{V}{H^3}=K_{\mathrm{frustum}}-K_{\mathrm{chamfers}}-K_{\mathrm{grooves}}
```

The components are:

Component	Non-dimensional expression	Meaning
Gross tapered square frustum	Kfrustum = (a2 + ab + b2) / 3	volume before groove and chamfer removals
Four corner chamfers	Kchamfers = 2d2	four triangular plan removals carried through the height
One circular groove	Ag = r2 arccos(e/r) - e sqrt(r2 - e2)	circular-segment area removed by one side groove
Four side grooves	Kgrooves = 4Ag	total groove subtraction
Final coefficient	KV = Kfrustum - Kchamfers - Kgrooves	final value of V/H3

Demonstration of the Square-Frustum Formula

Let a square frustum have height H, lower width A and upper width B. At a vertical coordinate z, measured from the bottom, the linearly interpolated side length is:

$$s(z)=A+\frac{B-A}{H}z$$

The horizontal area at level z is:

$$S(z)=s(z)^2$$

The frustum volume is therefore:

$$V_{\mathrm{frustum}}=\int_0^H\left(A+\frac{B-A}{H}z\right)^2\,dz$$

Expanding and integrating:

$$V_{\mathrm{frustum}}
=\int_0^H\left[A^2+2A\frac{B-A}{H}z+\left(\frac{B-A}{H}\right)^2z^2\right]dz$$

$$V_{\mathrm{frustum}}
=A^2H+A(B-A)H+\frac{(B-A)^2}{3}H$$

Collecting terms gives:

$$V_{\mathrm{frustum}}=\frac{H}{3}\left(A^2+AB+B^2\right)$$

Using $a = A/H$ and $b = B/H$:

$$\frac{V_{\mathrm{frustum}}}{H^3}=\frac{a^2+ab+b^2}{3}$$

This is the gross body used before subtracting the four lateral corner chamfers and the four circular side grooves.

Compact Numerical Balance

Convention	Gross frustum	Chamfers	Grooves	Final V/H3
Pita (1986)	1.093617000000	0.001152000000	0.067765000000	1.024700000000
Carvalho (2026)	1.067255782180	0.001124231482	0.066131550698	1.000000000000

For Pita, the one-groove area is:

$$A_g=\frac{1.092465000000-1.024700000000}{4}=0.016941250000$$

With r = 0.121500000000, this value is obtained by solving:

$$A_g=r^2\arccos\left(\frac{e}{r}\right)-e\sqrt{r^2-e^2}$$

which gives:

$$e=0.025906459610,\qquad q=0.118705961731,\qquad c_{\mathrm{arc}}=0.095593540390$$

For Carvalho, all Pita horizontal plan and groove coefficients are multiplied by:

$$f=\sqrt{\frac{1.000000000000}{1.024700000000}}=0.987874174182$$

Because the vertical reference height remains `H`, the non-dimensional area terms scale with f2, giving KV = 1.000000000000.

Dimensional Scaling from Input Weight

The script uses the input unit weight `W` and concrete unit weight `Wc` to compute the unit volume:

$$V=\frac{W}{W_c}$$

The selected dimensional convention then gives the reference height:

Convention	Height relation
Pita (1986)	H = (V / 1.024700000000)1/3
Carvalho (2026)	H = V1/3 = Dn

Once `H` is known, all absolute dimensions are obtained by multiplying the coefficient table values by `H`. For example, A = aH, B = bH, D = dH, R = rH and E = eH.

Nominal Diameter

The Van der Meer nominal diameter is:

$$D_n=V^{1/3}$$

The relation between Dn and the drawing height depends on the selected convention:

Convention	Relation
Pita (1986)	Dn = (1.024700000000)1/3 H = 1.008166460709H
Carvalho (2026)	Dn = H

This distinction is important when hydraulic sizing is expressed through Dn but CAD geometry is generated through a drawing height `H`.

Horizontal Section and Groove Construction

A horizontal section has local half-width h = w/(2H), where `w` is the non-dimensional section width. The east-side groove can be written in local non-dimensional coordinates as:

$$x(y)=h+e-\sqrt{r^2-y^2},\qquad -q\le y\le q$$

Key points are:

Feature	Coordinate or relation
Groove mouth	y = ±q, x = h
Deepest groove point	y = 0, x = h + e - r = h - carc
Other sides	obtained by 90-degree rotations
Corner chamfers	four lateral plan chamfers using offset d

This section definition is used consistently in IFC, STL, OBJ and DXF exports.

Hydraulic Stability Interpretation

Antifer hydraulic stability is not governed by geometry alone. It depends strongly on the armour-layer system: placement method, packing density, slope, wave climate, storm duration, underlayer, toe support, crest details and damage criterion.

The Hudson formula remains a common preliminary sizing relation:

$$W=\frac{\gamma_c H_D^3}{K_D\Delta^3\cot\alpha},\qquad \Delta=\frac{\gamma_c}{\gamma_w}-1$$

The stability number formulation is often more convenient for comparison:

$$N_s=\frac{H_s}{\Delta D_n}$$

For regular two-layer cubes and Antifer-calibrated formulae, a Van der Meer-type empirical structure may be used:

$$N_s=\left(k_1\frac{N_{od}^{k_2}}{N_z^{k_3}}+k_4\right)s_{0m}^{-k_5}$$

Typical coefficient sets are:

Armour unit and slope	k1	k2	k3	k4	k5
Regular cube, cot = 1.5	6.700	0.400	0.300	1.000	0.100
Regular cube, cot = 2.0	7.374	0.400	0.300	1.101	0.100
Antifer, cot = 1.5	6.951	0.443	0.291	1.082	0.082
Antifer, cot = 2.0	6.138	0.443	0.276	1.164	0.070

These formulae are preliminary design tools, not substitutes for project-specific verification. Reported KD values for Antifer vary widely because placement method and damage definition dominate the result.

Placement, Packing and Failure Modes

The same Antifer geometry can perform differently depending on armour-layer architecture.

Placement tendency	Main benefit	Main risk
regular placement	higher displacement stability	greater reflection and overtopping in some cases
irregular placement	greater roughness and reduced overtopping	lower displacement stability
dense packing	higher contact restraint	reduced porosity and possible paving behaviour
open packing	higher porosity and roughness	greater movement risk

Relevant failure and degradation mechanisms include:

Mechanism	Description
rocking	repeated angular movement without extraction
displacement	block moves out of its stable position
chain failure	movement of one block destabilises adjacent units
smoothing or paving	armour surface becomes more regular and less rough
concrete cracking	thermal, shrinkage, handling, impact or fatigue damage
toe or underlayer failure	loss of support leading to progressive armour instability
overtopping-induced rear damage	crest or rear-side damage due to transmitted water

A defensible Antifer design must therefore consider the full armour system, not only the isolated concrete unit.

Manufacture, Materials and Quality Control

Antifer units are normally manufactured as plain unreinforced concrete blocks. The main production advantages are simple formwork, robust geometry and straightforward inspection. The main risks are thermal cracking, honeycombing, density variation and handling damage.

Aspect	Typical practice or control
Concrete	normally unreinforced; low heat of hydration recommended
Strength class	commonly C25/30 in manual-type references
Formwork	four-face mould, open top and bottom in typical production
Casting	top casting with vibration

Aspect	Typical practice or control
Curing	prolonged moist curing, especially for large units
Handling	controlled lifting, rotation and storage procedures
Mass control	verify W = Wc V
Dimensional control	verify mould dimensions and generated volume
Rejection logic	reject significant cracks, honeycombing, cold joints or excessive corner loss

A dimensional error is a stability error. A density error is also a stability error. For CAD/BIM production, a volume error in the digital solid becomes a procurement, fabrication and quality-control error.

Python Script Installation and Usage

Requirements

Python 3.10 or newer is recommended. Tkinter must be available; on standard Windows Python installations it is normally included.

Install the runtime packages with:

```
python -m pip install ezdxf trimesh pillow
```

`pillow` is recommended for broad Windows and PyInstaller compatibility with optional `ezdxf` drawing add-ons.

Windows Virtual Environment

Create a virtual environment:

```
py -m venv .venv
```

Activate it:

```
.\.venv\Scripts\activate
```

Upgrade `pip`:

```
python -m pip install --upgrade pip
```

Install the dependencies:

```
python -m pip install ezdxf trimesh pillow
```

For standalone executable compilation, also install PyInstaller:

```
python -m pip install pyinstaller
```

Running the Application

Run the graphical application with:

```
python script_final_production.py
```

Typical workflow:

1. Enter the Antifer unit weight `W` in kN.
2. Enter the concrete unit weight `Wc` in kN/m3.
3. Select Pita (1986) or Carvalho (2026).
4. Select the output folder.
5. Review the calculation and validation preview.
6. Press **Generate IFC + STL + OBJ + DXF**.

Example output names:

```
antifer_pita1986_500kN_Wc24kNm3.ifc
antifer_pita1986_500kN_Wc24kNm3.stl
antifer_pita1986_500kN_Wc24kNm3.obj
antifer_pita1986_500kN_Wc24kNm3.dxf
output.txt
```

Viewing the Exported Files

Format	Recommended inspection
IFC	IFC/BIM viewers supporting faceted B-rep geometry
STL	mesh viewers or CAD applications with mesh import
OBJ	mesh viewers or modelling tools
DXF	AutoCAD/ZWCAD-type CAD inspection as ACIS-backed `3DSOLID`
TXT	direct review of dimensions and validation values

All exported geometry is dimensional and written in metres. The exported XY origin is placed at the homogeneous centre of mass, while the exported Z datum remains the physical bottom level z = 0.

Standalone Executable Compilation

A PyInstaller spec file is recommended for repeatable Windows production builds:

```
pyi-makespec --onefile --windowed --name script --collect-all ezdxf
↪  script_final_production.py
pyinstaller --clean --noconfirm script.spec
```

The compiled executable is written to the `dist` folder.

Verification Checklist

Before issuing geometry for technical use, check:

Check	Required result
selected convention	Pita or Carvalho coefficients match the master table
top face	one flat horizontal face at z = H, with no top bevel
chamfers	four lateral plan chamfers only

Check	Required result
grooves	one circular arc per side, radius R, centre offset E
closed shell	every B-rep edge shared by exactly two faces
volume	generated volume matches the selected KV
IFC/STL/OBJ/DXF	all formats generated from the same B-rep source
output report	dimensions and validation values recorded in `output.txt`

Limitations

This repository does not provide:

- new hydraulic model tests;
- prototype monitoring data;
- finite-element structural-integrity analysis;
- a complete breakwater design method;
- a universal Antifer KD;
- project-specific wave, overtopping, toe, crest or underlayer verification.

Critical works should be verified through physical modelling or equivalent project-specific evidence.

Conclusions

The production geometry is a reproducible algebraic reconstruction of an Antifer cube as a square frustum with four lateral plan chamfers and four circular side grooves. Pita preserves the historical volume coefficient V/H3 = 1.024700000000. Carvalho keeps the same topology but horizontally normalises the plan and groove coefficients so that V/H3 = 1.000000000000 and H = Dn.

The geometric definition is necessary for reliable CAD/BIM production, drawing control, volume calculation and quality assurance. It does not remove the need for hydraulic verification. Antifer stability remains governed primarily by armour-layer placement, packing density, wave climate, slope, storm duration, toe support, construction tolerance and accepted damage.

Dedication

This work is respectfully dedicated to the memory of **Carlos Alberto Roldao Maia Pita** (1950-2002), Portuguese civil engineer and specialist in maritime hydraulics and rubble-mound breakwater design. His LNEC *Memoria n.º 670 - Dimensionamento Hidraulico do Manto Resistente de Quebra-Mares de Talude* remains an important technical reference for Portuguese maritime engineering and provides a key historical basis for the geometric interpretation of Antifer cubes.

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